

Apply the Gauss elimination code to find the solution of the following systems:

$$\begin{array}{lll}
 w + 2x - 12y + 8z = 27 & 1.7x_1 + 2.3x_2 - 1.5x_3 = 2.35 & x_1 + x_2 + x_3 + x_4 + x_5 = 1 \\
 5w + 4x + 7y - 2z = 4 & 1.1x_1 + 1.6x_2 - 1.9x_3 = -.94 & x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 0 \\
 -3w + 7x + 9y + 5z = 11 & 2.7x_1 - 2.2x_2 + 1.5x_3 = 2.70 & x_1 + 3x_2 + 6x_3 + 10x_4 + 15x_5 = 0 \\
 6w - 12x - 8y + 3z = 49 & & x_1 + 4x_2 + 10x_3 + 20x_4 + 35x_5 = 0 \\
 & & x_1 + 5x_2 + 15x_3 + 35x_4 + 70x_5 = 0
 \end{array}$$

A farmer wants to prepare a feed formula for fattening cattle. He has at his disposal corn, waste, alfalfa and barley, each with certain units of nutritive ingredients, according to the following table:

FOOD					
Nutritional Ingredient	Corn	Food Waste	Alfalfa	Barley	Daily requirement Units/Kg
Carbohydrates	80	15	35	60	230
Proteins	28	72	57	25	180
Vitamins	20	20	12	20	160
Cellulose	50	10	20	60	80
Cost	18	5	7	20	-----

- Determine the kilograms of each material needed to meet the daily requirement (shown in the last column).
- Determine the cost of the food formula

Note: The food formula must contain all four foods.

All the exercise must be solved using Gauss method, remember that you received the code in the Word document. Follow the instructions in the extended classroom.